

CM500325 - Problem Sheet 1**Week 4 - Deadline 29 July 2026**

This is an individual assignment, contributing 50% of the final mark. An important part of mathematics is presenting your responses in a clear and unambiguous fashion, typeset, so that it is easy for others to read, understand and also assess them. It should be feasible to reach an acceptable standard of presentation using e.g. LaTeX, Microsoft Word, or LibreOffice Writer. To guarantee the anonymity of submissions, please avoid submitting handwritten answers, and avoid inserting front pages.

It is entirely possible to answer all responses, in a legible and clearly organised way, producing a pdf file with not more than 10 pages: please consider this as a general guideline so to maintain the focus on the questions. Please remember to reference all tools and sources that you use for your responses, including for typesetting your responses.

In relation to the use of Generative AI means, this assignment is categorised as Type B: the use of Generative AI tools is permitted, limitedly, just as assistive means only (for example for spelling), and yet it is not mandatory, nor necessary, to complete this assignment. As any other tools, they should be fully referenced.

Question 1 [9]: Translate the following sentences into propositional logic. Your formalizations should be as detailed as possible.

- a) Three is a prime number and is not an even number
- b) “If the speed limit is 30mph and I am driving at 25mph, then I am not breaking the law” .
- c) “Bob will go to the grocery shop only if the fridge is empty” .

Question 2 [9]: Translate the following sentences into predicate logic. Your formalizations should be as detailed as possible.

- a) Fairies do not exist.
- b) Somebody has visited all the concert halls in London
- c) Not Everyone has a sibling.

Question 3 [9]: Consider the following formula F of propositional logic:

$$((R \wedge I) \vee (\neg S \vee \neg P)) \rightarrow \neg C$$

- a) Suppose R = “the router is on” , I = “internet is available” , S = “signal is strong” , P = “entered password is correct” , C = “the device is connected” . Translate F in an english sentence.
- b) Which situation is modelled by that sentence, where the propositions are interpreted as described?

- c) Assume that R, I, C and S are true, and P is false. Is F true or false? Briefly explain why.

Question 4 [8]: Consider this sentence: ‘Every device that can reach a router, and knows the password of that router, will connect to it.’ This sentence is formalised, in predicate logic, with P :

$$P = \forall d \forall r ((Can - reach(d, r) \wedge Knows - pwd(d, r)) \rightarrow Connected(d, r)).$$

Besides, consider the formula Q :

$$Q = \forall d (\exists r (Can - reach(d, r) \wedge Knows - pwd(d, r)) \rightarrow \exists r Connected(d, r)).$$

1. In a domain with 2 devices and 3 routers, give a model where the formula P is falsified.
2. Explain whether Q provides a different formalisation than P given the initial sentence, and why.

Question 5 [16]: Identify which of these relations are reflexive which are symmetric, which are transitive, and which are equivalence relations. You do not need to show any working, but must clearly state which properties hold and which do not. Let S be a set, and consider a relation $R \subseteq S \times S$.

1. A is the set of all animals, $R_1 = \{(a, b) \mid a \text{ is the same species as } b\} \subseteq A \times A$
2. $R_2 = \{(m, n) \mid m^3 \leq n^3\} \subseteq \mathbb{Z} \times \mathbb{Z}$
3. $R_3 = \{(x, y) \mid x + y < 1\} \subseteq \mathbb{R} \times \mathbb{R}$
4. $S = \{a, b, c\}$, $R_4 = \{(a, b), (b, c), (c, a)\} \subseteq S \times S$.

Question 6 [15]: Let $f: \mathbb{N} \rightarrow \mathbb{Z}$ be the function defined as

$$f(x) = \begin{cases} x/2, & \text{if } x \text{ is even} \\ -\frac{x+1}{2}, & \text{if } x \text{ is odd} \end{cases}$$

1. Demonstrate that f is injective.
2. Prove that f is a bijection.
3. Find the inverse function $f^{-1}: \mathbb{Z} \rightarrow \mathbb{N}$

Question 7 [16]: Consider the functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ defined as follows:

$$f(x) = \sin(5x + \pi) \quad g(x) = \sin(2x)e^{2x}.$$

Answer the following questions and show some working before the final answer.

1. Find $f'(x)$ and $g''(3)$
2. Calculate $\int f(x)dx$ and $\int_0^{\pi/4} g(x)dx$
3. Is the function $g(x)$ differentiable in all points in $\mathbb{R}$? Justify your reasoning.
4. What is the behaviour of $g(x)$ for $x \rightarrow \infty$?

Question 8 [18]: A square matrix $(n \times n)$ is said to be in upper(/lower) triangular form when all values below(/above) the main diagonal are zero. Formally, if $M = (m_{ij})$ is an $n \times n$ matrix, we say that M is upper triangular if $m_{ij} = 0$ when $i > j$. (A comparable definition holds for lower triangular ones.) For example, the matrix A is an upper triangular one:

$$A = \begin{pmatrix} 5 & 1 & -3 \\ 0 & 2 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

Besides, on any square matrix N , its transpose matrix N^T is obtained by rewriting the rows of N as columns in N^T , preserving the order (first row becomes first column,...)

1. Formally prove that for an upper triangular matrix M , its transpose M^T is a lower triangular matrix.
2. Compute A^{-1} , its eigenvalues and eigenvectors, and show some processing before the end result.
3. Doing some own research, explain how triangular matrices can be used in graphics to capture scale and shear information (it is sufficient to refer to the 2D case).